5.3 DENSITY ALTITUDE

The density altitude chart (see Fig. 5-3) expresses density altitude in terms of pressure altitude and temperature. The less dense the air the higher the density altitude. For standard conditions of temperature and pressure, density altitude is identical to pressure altitude.

A high density altitude affects the performance of both rotors and engines. When density altitude is high, less lift is developed by the rotor blades for any given power setting than at standard conditions and the power output of the engines is reduced below the output for standard conditions.

Each takeoff and landing must be separately evaluated as density altitude may change considerably in a short period of time.

The true airspeed factor value $\frac{1}{\sqrt{\sigma}}$ is a conversion factor used to obtain true airspeed from calibrated airspeed by correcting for density altitude.

<u>EXAMPLE</u>: (see Fig. 5-3)

Determine: **Density altitude (DA)**

True airspeed factor

True airspeed (TAS)

Known:	OAT	–14	°C
	Pressure altitude	5000	ft
	CAS	100	kt

Solution:
1. Enter chart at known OAT (–14 °C).
2. Move up to known pressure altitude (5000 ft).
3. Move left and read density altitude = <u>2630 ft</u>.
4. Move right and read true airspeed factor = <u>1.04</u>.
5. Multiply the known calibrated airspeed (100 kt) by true airspeed factor ($\frac{1}{\sqrt{\sigma}}$ = 1.04) to obtain true airspeed.

   TAS = CAS x $\frac{1}{\sqrt{\sigma}}$ = 100 x 1.04 = <u>104 KTAS</u>

DENSITY ALTITUDE CHART

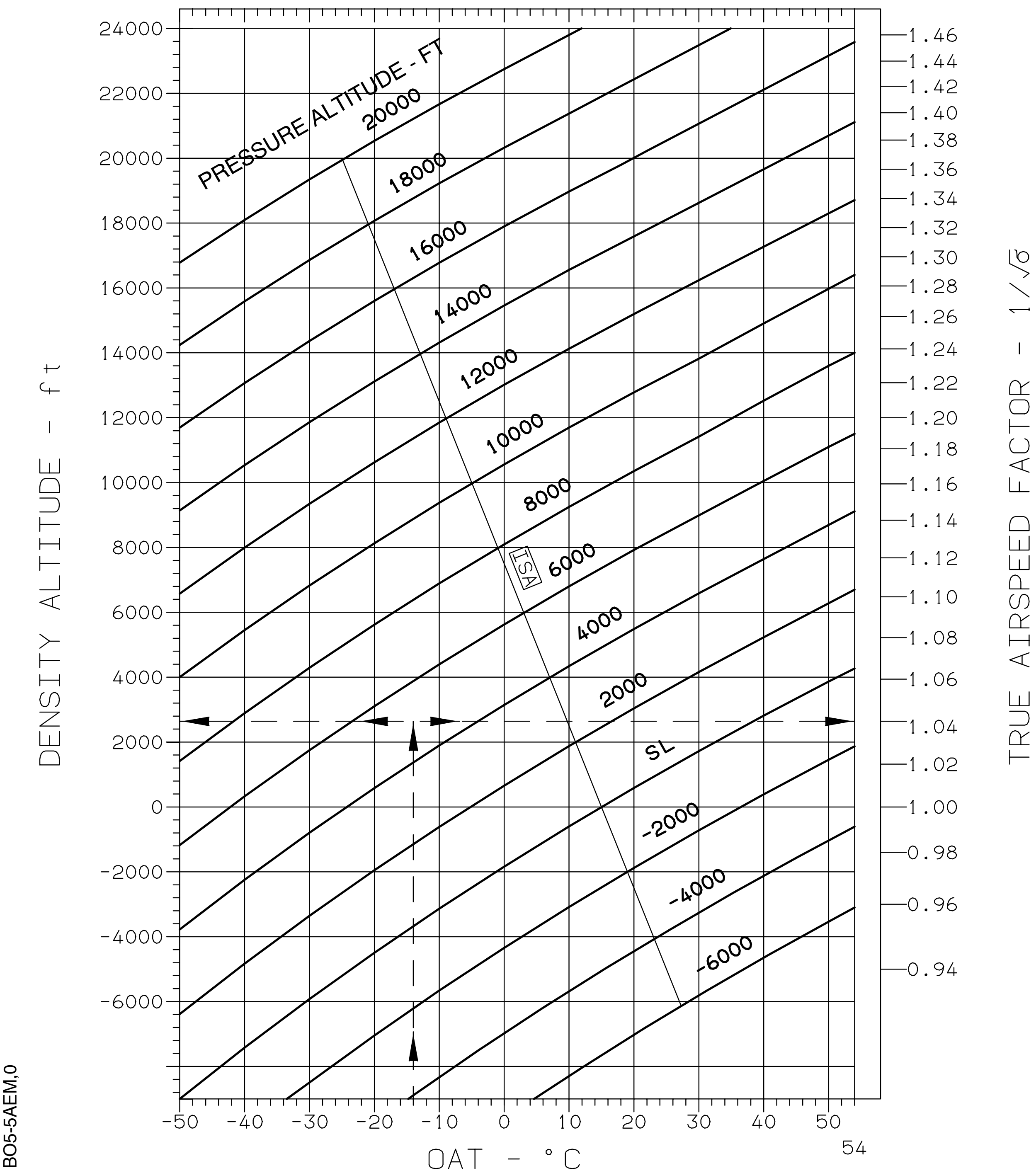

Fig. 5-3 Density Altitude Chart